

Paper Practice 2.3 • WORKED KEY**Solving Multi-Step Equations**

WORKED KEY (odd-numbered items) – Undo addition/subtraction first, then multiplication/division. Clear a fraction by multiplying both sides by its denominator. Even-numbered items are yours to work.

Use properties of equality to solve each equation. Check your solution.

1. $5c + 3 = 18$

Answer: $5c = 15 \rightarrow c = 3$. Check: $5(3) + 3 = 18$ ✓

2. $9w + 8 = -1$

3. $43 = 8z + 11$

Answer: $32 = 8z \rightarrow z = 4$. Check: $8(4) + 11 = 43$ ✓

4. $\frac{y}{3} + 5 = 14$

5. $\frac{x}{4} + 18 = 9$

Answer: $\frac{x}{4} = -9 \rightarrow x = -9 \cdot 4 = -36$. Check: $-9 + 18 = 9$ ✓

6. $\frac{h}{-3} + 7 = 8$

7. $12 = \frac{4 - a}{2}$

Answer: $24 = 4 - a \rightarrow a = 4 - 24 = -20$. Check: $(4 + 20) \div 2 = 12$ ✓

8. $\frac{5 + m}{6} = 4$

9. $\frac{18 - p}{-4} = -4$

Answer: $18 - p = (-4)(-4) = 16 \rightarrow p = 2$. Check: $16 \div (-4) = -4$ ✓

10. **PHOTOGRAPHY** A photographer sells his photographs for \$20 each at an art show. He spent \$185 to print the photographs and made a profit of \$55 at the art show. Write and solve an equation to find p , the number of photographs sold.

11. **BOWLING** Brooke went bowling with her friends. She bowled 3 games, and her average score for those games was 121. She had 117 her first game and 108 her second game. Write and solve an equation to find g , the score of her third game.

Answer: $\frac{117 + 108 + g}{3} = 121 \rightarrow 225 + g = 363 \rightarrow g = 138$. Check: $(117 + 108 + 138) \div 3 = 363 \div 3 = 121$ ✓

Solve each equation for x . Assume $a \neq 0$.

12. $ax + 5 = 13$

13. $ax - 9 = 15$

Answer: $ax = 24 \rightarrow x = \frac{24}{a}$

14. $7 = 4 - ax$

15. $3 = \frac{5}{ax} + 5$

Answer: $-2 = \frac{5}{ax} \rightarrow -2ax = 5 \rightarrow x = -\frac{5}{2a}$